

Bharatiya Vidya Bhavan's
SARDAR PATEL INSTITUTE OF TECHNOLOGY
(An Autonomous Institute Affiliated to University of Mumbai)
MUNSHI NAGAR, ANDHERI (WEST), MUMBAI - 400058
MID SEMESTER EXAMINATION

Sr.No. **8515**

L-29, 486, 986, 1, 1505, 38373

AY:	2025-26	Session:	Ddd MSE October 2025	Date:	08-10-2025
Sem:	1	Slot:	10:00am - 11:00am	C. Code:	CS413
Exam Type:	Written	Block:	401	Seat No:	2022600029
		Course Section:	1		



Name of the candidate. Mayur Madhukar Mahajan

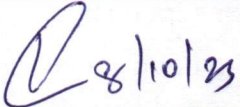
Candidate's UID number:

2	0	2	2	6	0	0	0	2	9
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Examination class room number:

4	0	1
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Invigilator's signature with date:



(To be filled in by the candidate)
EXAMINATION: BE BTech MSE
(For example FY MCA/FY BTECH)
PROGRAM: CSE(AI/ML)
DAY AND DATE: Wednesday 08-10-2025
COURSE NAME: Deep Learning

SEMESTER : VII
I / II / III / IV / V / VI / VII / VIII

(To be filled in by the examiner)

Question numbers	1	2	3	4	5	6	Total	Signature of the examiner
Marks awarded								
Marks after revaluation								

INSTRUCTIONS TO CANDIDATES

- 1) Candidate should ensure that all answer books including supplementary answers books received by them bear the signature of the junior supervisor, otherwise the answer books will not be examined.
- 2) Begin the answer to each question on a new page. For each answer, write the corresponding question number and sub question number if any in the respective column provided in the answer sheet.
- 3) Do not write anything in the column provided for the marks to be assigned by the examiners.
- 4) Candidate will not be permitted to leave the examination hall until an hour after the question papers are distributed.
- 5) Every candidate present must sign against their seat number on the attendance sheet provided by the junior supervisor.
- 6) Candidates are forbidden to (i) bring any book, notes. Scribbling papers, mobile telephones, programmable calculators, or any other similar devices.(ii) speak or communicate in any manner to any other candidate, while the examination is in progress, and (iii) take with them any answer-book written or blank while leaving the examination hall. The supervisors/authorized persons are authorized to check the students.
- 7) Candidate suspected to be guilty of any of the aforesaid acts will be allowed to write their paper only after giving an undertaking in writing that the decision of the College Examination Authorities in respect of the reported act of unfair means is binding on them.
- 8) Candidates should write their answers legibly. They are warned that **zero marks** will be assigned to **answers which cannot be assessed by the examiners owing to illegible handwriting**.
- 9) Write on both side of a page. Rough work, when necessary, should be done on the Left- hand side and in pencil only.
- 10) While underlining of answers for focusing attention is permitted, use of varied inks, except for illustration and figures must be avoided.
- 11) No sheet shall be torn from the answer-books provided nor shall additional papers attached to them.
- 12) The answer-books will be scrutinized before they are sent to examiners.
- 13) All answer-books supplied shall be returned whether written or blank.
- 14) Nothing shall be written on the question-paper.
- 15) If candidate needs anything, they should approach the Junior Supervisor without disturbing other candidates. However, they should not leave their seat on my account.
- 16) A warning bell will be given ten minutes before the close of the examination. Candidates will not be allowed to leave the examination hall during the last ten minutes. At the final bell, they must stop writing and be ready to hand over their answer-books to the Junior Supervisor. They should not leave their seats until answers-books from all candidates are collected by the Junior Supervisor.
- 17) A candidate who disobeys any instructions issued by the Senior/Junior Supervisor or who is guilty of rude or disobedient behavior is liable for disciplinary action to be taken against him/her by the College Examination Authorities.



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Question No.

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$$f(w) = \frac{1}{2} \|xw - y\|^2$$

Q. 1a

$$x_{t+1} = x_t - \alpha f'(t) \text{ is a gradient descent}$$

$$f'(x) = \frac{1}{2} \|xw - y\|^2 (x - 0)$$

$$= \frac{1}{2} \cdot x \cdot \|xw - y\|^2$$



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Q.1 b)

$$SSR = \sum_{i=1}^n (\text{predicted}_i - \text{output}_i)^2 = \sum_{i=1}^n y_i^2$$

$$\frac{d(SSR)}{d(\text{predicted})} = \sum_{i=1}^n \frac{d'(\text{predicted}_i - \text{output}_i)^2}{d(\text{predicted})}$$

$$= \sum_{i=1}^n 2(\text{predicted} - \text{output}) \times (1 - 0)$$

$$= \sum_{i=1}^n 2 \times y_i$$

$$= 2 \sum_{i=1}^n y_i$$


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Q3d In RNN, due to long sequential input of previous context h_{t-1} , vanishing gradient problem arises, whereas in LSTM, forget gate and input gate contributes for forgetting unnecessary information and adding required information to the the neural network respectively..

equation for forget gate:

$$f_t = \sigma(W_f[h_{t-1}, x_t] + b_f)$$

equation for input gate:

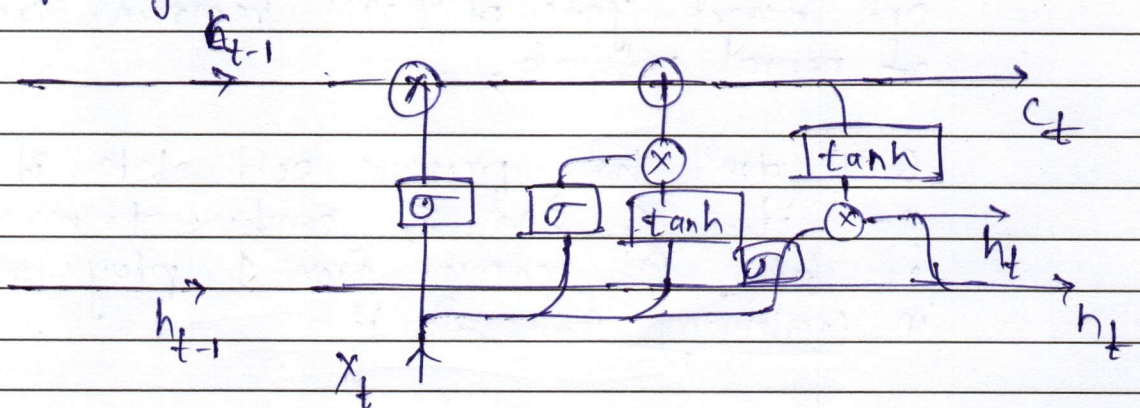
$$i_t = i_t \otimes \tilde{c}_t$$

$$i_t = \sigma(W_i[h_{t-1}, x_t] + b_i)$$

$$\tilde{c}_t = \tanh(W_c[h_{t-1}, x_t] + b_c)$$

This f_t function output is bitwise multiplied with h_{t-1} to decide what information should be forgotten and which to include in neural network

And again, vital information is added by input gate to neural network.



$$c_t = f_t \otimes c_{t-1}$$

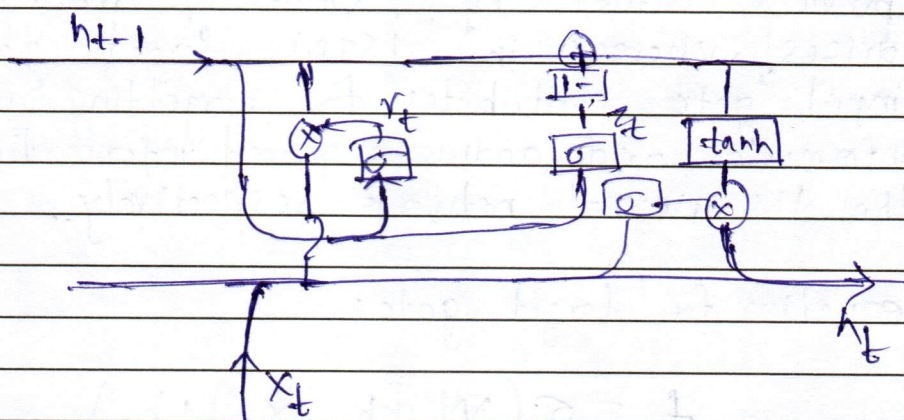
$$h_t = \tanh(i_t \otimes \tilde{c}_t + c_t) \otimes h_{t-1}$$

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Q3b) GRU:



This is an architecture of gated recurrent unit.

It has reset gate

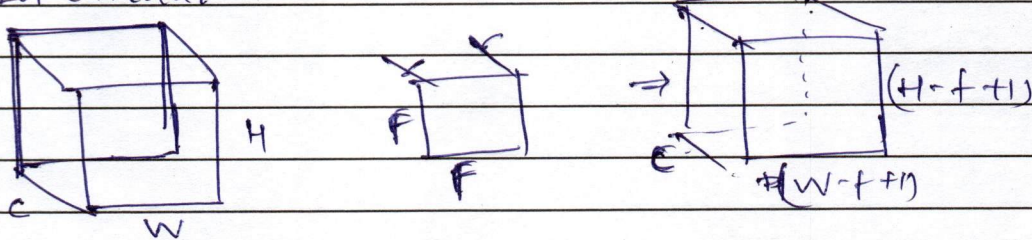
$$r_t = \sigma(W_r[h_{t-1}, x_t] + b_r)$$

$$z_t = \sigma(W_z[h_{t-1}, x_t] + b_z)$$

$$h_t = r_t \otimes (1 - z_t) \oplus h_{t-1}$$

It has less parameters than LSTM, ~~this is~~ This reset gate works as input gate and forget gate of LSTM, removing complexity of neural network.

GRU don't has previous cell state, it doesn't need it. r_t and z_t combine to make h_t of GRU. Bit removal from 1, plays role in combining rules for it.

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	Q2a)	2d convolution 

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Sr. No.

8515



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Question
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Sr. No. 8515



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